Prompt Sensitivity in Zero-Shot CLIP: How Much Does Wording Alone Move Accuracy on Look-Alike Traffic Signs?

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Code, instructions for reproducing the experiments, and all results are available at:
`https://github.com/Shivash30/wits-cv-prompt-sensitivity`

***Abstract*— Zero-shot vision–language models such as CLIP classify an image by comparing it with one written description per class. We measure how much top-1 accuracy changes when only the wording of that description changes. With CLIP ViT-B/32, 5,880 GTSRB test photographs of 18 visually similar traffic-sign classes, and seven fixed, meaning-preserving prompt templates, accuracy ranged from 42.6% to 53.5%: a spread of 10.9 percentage points (pp), 95% bootstrap interval 9.9–12.1 pp. This exceeds the 10 pp threshold proposed in our hypothesis, although the interval includes values slightly below it. Every template differed significantly from the standard "a photo of a {class}" baseline (McNemar $p < 10^{-12}$), the bare class name beat that baseline by 3.5 pp, and an irrelevant phrase ("taken on a sunny day") cost 3.4 pp. Per class, the effect is far larger: half of the classes move by more than 45 pp between templates. CLIP's own confidence was a poor guide: it ranked the templates almost in reverse order of accuracy (Spearman $\rho = -0.61$).**

***Index Terms*— CLIP, zero-shot classification, prompt sensitivity, traffic signs, calibration**

I. Problem and What We Measured

CLIP [1] classifies an image without task-specific training: each class is written as a sentence, the image and the sentences are embedded, and the closest sentence wins. The sentence is built from a *template* such as "a photo of a {class}". Two templates that describe the same class should, ideally, give the same answer. If they do not, then the accuracy reported for a zero-shot model is partly a property of the wording somebody happened to choose. The same issue arises in any deployed system that matches inputs to written labels; in a contact-centre chatbot, for example, the wording of each intent's sample utterances determines where a caller is routed.

Question. With the model, images and classes held fixed, how much does top-1 accuracy change when only the prompt template changes?

Hypothesis (stated in our proposal before any run). The spread between the best and worst of seven meaning-preserving templates exceeds 10 pp. We chose 10 pp in advance as a difference large enough to change which prompt, or which model, a practitioner would choose; it is a judgement about practical importance, not a statistical constant.

We chose a fine-grained task, traffic-sign recognition with deliberately similar classes, because that is where we expected wording to matter most. We also report an analysis not in the proposal: whether CLIP's confidence could be used to choose a template without labelled data.

II. Data

Source. We used the German Traffic Sign Recognition Benchmark (GTSRB) [2]: real-world photographs of road signs captured from a moving vehicle, labelled by hand by the dataset's creators into 43 classes. The official download host timed out from Google Colab, so we used a Kaggle mirror of the official files (`meowmeowmeowmeowmeow/gtsrb-german-traffic-sign`), which ships the official `Test.csv` labels. Our code checks the mirror against the official test split before running (12,630 images in total, and the exact image counts of six reference classes) and records the source used in `results/data_source.txt`.

Subset and size. We kept 18 classes chosen for visual similarity: eight speed limits (20, 30, 50, 60, 70, 80, 100 and 120 km/h), which differ only in the printed number, and ten red warning triangles (GTSRB ids 18–27), which differ only in the small pictogram. This gives 5,880 test images. The classes are unbalanced, from 60 images (e.g. 20 km/h, pedestrians) to 750 (50 km/h); per-class counts are in `results/per_class_accuracy.csv`.

Split. No model was trained or fine-tuned, so only the official test split was used, once. No image was used to choose or adjust the templates, which were fixed in the proposal.

Licence and consent. We used a Kaggle mirror containing the official GTSRB files and labels. The mirror lists the data under the CC0 1.0 (public-domain) dedication, which is also the position recorded in our proposal; we did not find a separate licence statement on the original GTSRB page, so we record this as the position we found rather than a verified licence of the original release. The images show road signs, not identifiable people, so no consent question arose.

III. Method

A. Zero-shot classification

For an image x and class prompts $t_1, \ldots, t_K$ ($K = 18$), CLIP's image encoder f and text encoder g produce unit-normalised embeddings. The score for class k and CLIP's class probabilities are

$$s_k = \tau \cos\big(f(x), g(t_k)\big), \qquad p_k = \frac{e^{s_k}}{\sum_j e^{s_j}}, \tag{1}$$

with CLIP's learned temperature $\tau = 100$. The prediction is $\arg\max_k s_k$ and its *confidence* is $\max_k p_k$. Because only the text changes between templates, each image is encoded once and re-scored against seven sets of 18 text embeddings.

B. Templates and class names

Table I lists the seven templates from our proposal. They vary in specificity, context and detail while naming the same class. T2 is the default used in the CLIP paper and is our baseline; T7 adds a phrase that is irrelevant to the sign's identity and acts as a control. Class names (e.g. "speed limit 30 km/h", "road narrows on the right") omit the word "sign" so that T5 reads grammatically; this choice is applied identically to every template.

C. What we reused and what we wrote

Reused, unchanged: the pretrained CLIP ViT-B/32 weights released by OpenAI [1], loaded through Hugging Face `transformers` [3]; the GTSRB images and labels; and standard libraries (PyTorch, NumPy, pandas, SciPy for the binomial test, matplotlib). No weights were changed and nothing was fine-tuned.

Changed in reused code: nothing in the libraries themselves. One implementation detail: instead of `get_image_features`/`get_text_features`, whose return type differs between `transformers` versions, we call CLIP's encoders and projection layers directly. This computes the same embeddings and keeps the code working across versions.

Written by us (`prompt_sensitivity.py`): the class selection and names; the templates; dataset loading with mirror fallback and label-count checks; caching of image embeddings; scoring of every image under every template; the statistics (bootstrap intervals, McNemar tests, calibration measures); the template ensemble; and every figure and results file.

IV. Experiment

Held constant. Model and weights, the 5,880 images, the 18 class names, image pre-processing (CLIP's own), the metric and the random seed. Only the template changes. All seven templates were fixed before the first run and each was evaluated exactly once on the same images, so no template received more tuning than the baseline; there were no hyper-parameters to tune.

Metric. The primary metric is top-1 accuracy, the fraction of images whose highest-scoring class is correct, as reported in the CLIP paper. Because the classes are unbalanced, we also report balanced accuracy (the mean of per-class accuracies). The headline quantity is the *spread*: the maximum minus the minimum top-1 accuracy across T1–T7, in percentage points.

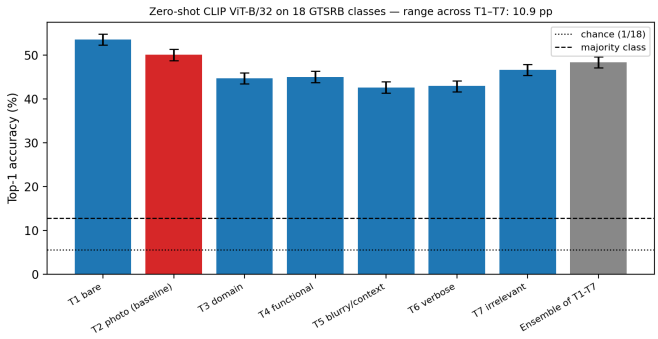


Figure 1: Top-1 accuracy (%) for each template with 95% bootstrap intervals. The baseline T2 is red and the ensemble grey; the dotted and dashed lines are chance (5.6%) and the majority-class predictor (12.8%).

Baseline and reference points. Each template is reported as a difference from T2. To show that CLIP is doing real work, we also give random guessing ($1/18 = 5.6\%$) and a majority-class predictor that always outputs the most frequent class (50 km/h; 12.8%).

Uncertainty. 95% intervals come from 1,000 bootstrap resamples of the images [4] (seed 0). The interval on the spread applies the same resample to all templates. Differences from T2 are tested with an exact McNemar test [5] on paired per-image correctness.

Confidence. For each template we compare mean confidence with accuracy, compute the expected calibration error (ECE, 15 bins) [6], and ask whether ranking the templates by mean confidence, which needs no labels, recovers their ranking by accuracy (Spearman correlation across the seven templates).

Ensemble. As an extra, we average the seven normalised text embeddings of each class and classify with the averaged embedding.

Reproducibility. The run takes about 10 minutes on a free Colab T4 GPU. CLIP inference is deterministic; the only randomness is bootstrap resampling, which is seeded. All numbers below are in `results/template_accuracy.csv`, `spread_summary.json`, `per_class_accuracy.csv` and `confusion_*_template.csv` in the repository.

V. Results

A. Headline: a 10.9 pp spread

Table II and Fig. 1 give the main result. Top-1 accuracy ranged from 42.6% (T5) to 53.5% (T1), a spread of **10.9 pp** (95% interval 9.9–12.1 pp; mean 46.5%, standard deviation 4.0 pp across templates). The observed spread exceeds the proposed 10 pp threshold, although the confidence interval includes values slightly below that threshold. In balanced accuracy the spread is 9.9 pp (31.1% to 41.0%), just under the threshold. All templates are far above both reference points.

Three findings hold regardless of where the threshold is set.

Table I: The seven prompt templates, shown for the 30 km/h class.

ID	Style	Example
T1	Bare class name	`speed limit 30 km/h`
T2	CLIP default (baseline)	`a photo of a speed limit 30 km/h`
T3	Domain-qualified	`a photo of a speed limit 30 km/h, a type of traffic sign`
T4	Functional	`a road sign indicating speed limit 30 km/h`
T5	Degraded / context	`a blurry photo of a speed limit 30 km/h sign on a road`
T6	Verbose	`a photograph of a speed limit 30 km/h, showing its visual characteristics and function as a traffic sign`
T7	Irrelevant detail (control)	`a photo of a speed limit 30 km/h, taken on a sunny day`

Table II: Accuracy per template on 5,880 images (%). Δ is the difference from T2 in pp; every Δ has McNemar $p < 10^{-12}$ for T1–T7 ($p = 8 \times 10^{-5}$ for the ensemble). Conf. is CLIP's mean confidence in its prediction.

Template	Top-1 [95% CI]	Δ (pp)	Balanced	Conf.	ECE
T1 bare	**53.5** [52.3, 54.7]	+3.5	41.0	28.4	0.26
T2 baseline	50.0 [48.7, 51.3]	—	36.6	28.9	0.24
T3 domain	44.6 [43.5, 45.9]	−5.4	35.0	29.0	0.16
T4 functional	45.0 [43.7, 46.3]	−5.0	36.1	30.0	0.17
T5 blurry/context	42.6 [41.3, 43.9]	−7.4	32.1	31.1	0.14
T6 verbose	42.9 [41.6, 44.1]	−7.1	31.8	30.1	0.13
T7 irrelevant	46.6 [45.4, 47.9]	−3.4	31.1	32.9	0.16
Ensemble T1–T7	48.3 [47.0, 49.6]	−1.8	35.7	32.3	0.16
Chance	5.6				
Majority class	12.8				

First, every template differs from the baseline by more than chance would explain. Second, the CLIP default is not the best wording here: the bare class name (T1) beat it by 3.5 pp. Third, the irrelevant "taken on a sunny day" (T7) still cost 3.4 pp. Averaging all seven templates (48.3%) did not beat the baseline.

B. *Per class, the effect is much larger*

Fig. 2 shows that the aggregate hides large movements that partly cancel. Nine of the 18 classes have a per-class range above 45 pp. "Pedestrians" falls from 83.3% (T5) to 6.7% (T7) when "taken on a sunny day" is added; "general caution" rises from 15.9% (T1) to 86.2% (T6); "speed limit 30 km/h" falls from 76.3% (T1) to 10.6% (T6). Apart from classes that are near 0% under every template (Section VI), only 70 km/h (range 4.1 pp), 100 km/h and 50 km/h (about 10 pp each) are stable.

C. *Confidence does not identify the best template*

CLIP was *under*-confident under every template: mean confidence of 28–33% against accuracy of 43–54% (Fig. 3; ECE 0.13–0.26). More importantly for practice, mean confidence ranked the templates almost in reverse order of accuracy (Spearman $\rho = -0.61$). The most confident template was T7, at 46.6% accuracy, not T1. A practitioner choosing a template by confidence alone, without labelled data, would have chosen a worse one.

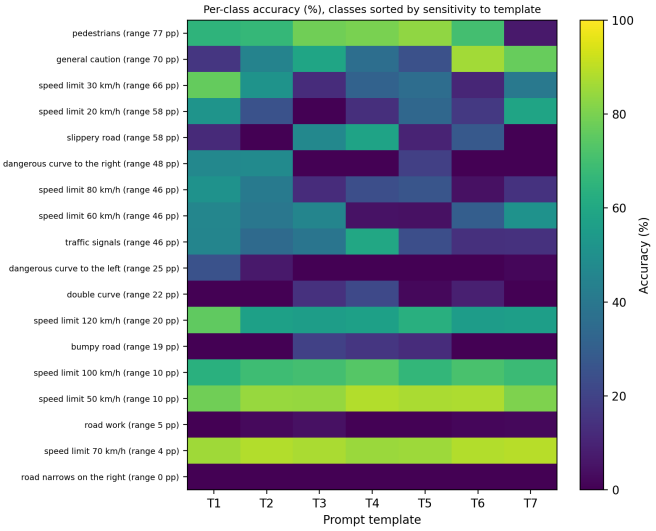


Figure 2: Per-class top-1 accuracy (%) for each template. Rows are sorted by their range across templates (pp, in brackets).

VI. Failures and Limitations

Where the method breaks. The confusion matrices for T1 and T5 (`results/confusion_*_template.csv`) show four recurring failures:

- *Some classes are never recognised.* "Road narrows on the right" is 0% under every template; under T1 it is split between the two "dangerous curve" classes (32% and 31% of its images), and under T5 84% of it is labelled "pedestrians". "Road work" never exceeds 4.6%.

- *An attractor class.* Under T5, "pedestrians" absorbs most warning triangles: 74% of road-work, 67% of slippery-road, 57% of general-caution and 56% of traffic-signal images.

- *Digits are poorly read.* Under T5, 65% of 60 km/h, 42% of 80 km/h and 39% of 30 km/h signs are labelled 50 km/h; even under T1, 23% of 100 km/h signs are labelled 120 km/h.

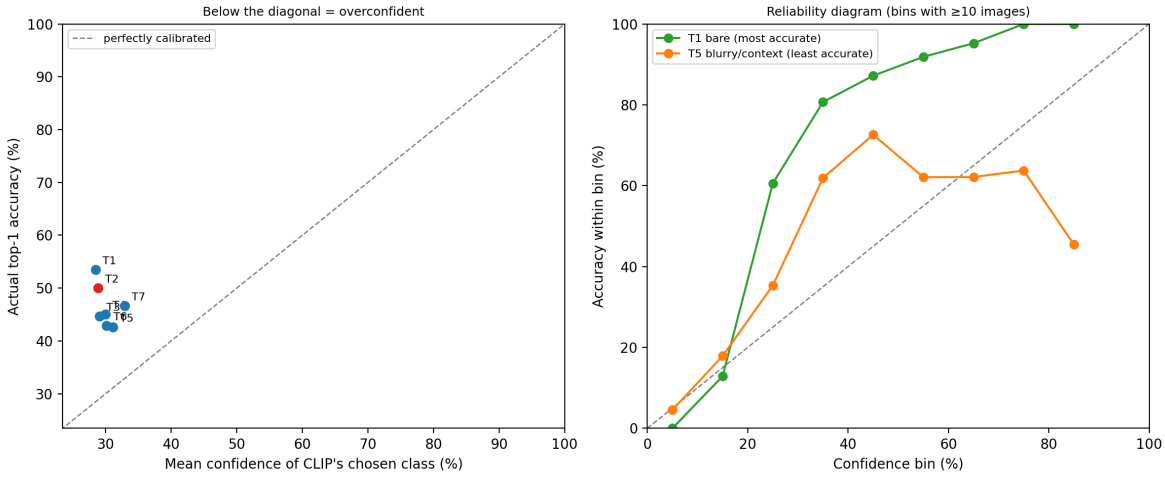


Figure 3: Left: mean confidence of the predicted class (%) against top-1 accuracy (%), one point per template; points above the diagonal are under-confident. Right: reliability diagram for the most (T1) and least (T5) accurate templates, accuracy (%) within each 10% confidence bin with at least 10 images.

- *Left and right are confused.* Under T1, 65% of "dangerous curve to the left" images are labelled "to the right".

In each case wording changes *which* wrong answer CLIP gives, but cannot fix a visual distinction the model does not make.

What did not work as expected. The observed spread exceeds the 10 pp threshold, but its confidence interval reaches slightly below it, and on balanced accuracy the spread (9.9 pp) falls just under it. We expected the template ensemble to be at least as good as the baseline; it was 1.8 pp worse.

Limitations. (i) One model (ViT-B/32) and one dataset; larger CLIP models may be less sensitive. (ii) Seven handwritten templates are a small sample of possible wordings, and they differ from one another in several ways at once, so we cannot attribute an effect to a single feature (e.g. T5 changes both the scene description and the word "sign"). (iii) Class names are themselves a wording choice that we held fixed rather than varied. (iv) Images were used uncropped, without GTSRB's sign bounding boxes. (v) German signs only.

VII. What We Would Do Next

Repeat the experiment with larger CLIP models and a few hundred automatically generated templates, so that the spread can be estimated as a distribution rather than from seven points, and vary class names as well as templates. We would also test whether learned prompts [7] remove the sensitivity, and run the same protocol on a production intent classifier with labelled utterances.

VIII. Use of AI Tools

We used Claude (Anthropic), an AI assistant, to draft the experiment code from our proposal, to diagnose run-time problems (a `transformers` API change and an unreachable dataset host), and to draft the presentation and the text of this report. We ran every experiment ourselves on Colab, checked the reported numbers against the files in the repository, and edited the text. We are responsible for all code and claims.

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